

Pelvic Geometry Dominates Load-Dependent Sacroiliac Joint Micromotion

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Abstract

Background: Sacroiliac joint (SIJ) micromotion depends on pelvic anatomy, material properties, and load direction, yet large-cohort evidence under standardized standing and parturition-motivated loading remains limited, particularly for separating geometric versus material sources of variability.

Methods: We analyzed a precomputed finite-element repository of 278 pelves (150 female, 128 male; age 16–91 years) represented on a common template via an implicit subject-specific mapping. Five standardized static load cases were studied: two-leg standing (SP2leg), one-leg standing (SP1leg), and three parturition-motivated proxies (LAB1–LAB3). Outcomes were SIJ rotation/translation magnitudes and AP/ML/CC translation components. We used robust and cluster-robust regression, variance-channel ratios (shape-only/full and material-only/full), and log–log size-scaling models against total pelvic volume.

Results: SP1leg produced the largest SIJ motion (median rotation 2.52°; translation 1.76 mm). Relative to SP2leg, LAB loads reduced translation magnitude but showed clear directional re-routing of AP/ML/CC components. Female sex was associated with higher adjusted LAB translation magnitude (~0.2–0.23 mm) and higher LAB3 rotation. Between-subject variability was almost entirely geometric (shape/full 98.7–102.1%), while material-only variance fractions were small (0.001–0.317%). Larger pelvic volume tended to predict lower standing micromotion, with no robust sex-by-size slope interaction after false-discovery

correction.

Conclusions: Under standardized forces, SIJ response is dominated by load-dependent directional re-routing and pelvic geometry, while material mapping contributes little to between-subject spread. Size scaling is load dependent and generally negative under standing loads.

Keywords: sacroiliac joint; pelvic anatomy; finite-element analysis; sexual dimorphism; scaling; parturition biomechanics

1 Introduction

The sacroiliac joint (SIJ) is a mechanically demanding articulation of the pelvic ring: it must transmit large forces between the trunk and the lower limbs while allowing only small, tightly constrained micromotions (Vleeming et al., 2012; Snijders et al., 1993). That combination—high load transfer with low mobility—is achieved through irregular articular surfaces, strong ligamentous restraint, and the closed-ring geometry of the pelvis (Vleeming et al., 2012; Snijders et al., 1993).

Because these motions are small, they are challenging to measure consistently in vivo and across experimental set-ups. Reported SIJ kinematics are typically sub-degree to low-degree rotations with sub-millimetric to low-millimetric translations, and they depend strongly on both task and measurement approach (Goode et al., 2008; Meijer et al., 2014; Keizer et al., 2019). Crucially, SIJ motion should not be expected to scale uniformly across directions: translation components in the anteroposterior (AP), mediolateral (ML), and craniocaudal (CC) axes can change disproportionately with loading direction and constraint, reflecting anisotropic joint anatomy and the ligamentous network of the pelvic ring (Vleeming et al., 2012; Goode et al., 2008; Meijer et al., 2014).

Directional behavior becomes most informative when loading changes. Unilateral support in locomotion increases asymmetry compared with quiet two-leg standing, and instrumented studies show that routine activities can generate large, task-dependent hip forces that must be routed through the pelvic ring (Bergmann et al., 2001; Kutzner et al., 2016). In obstetric contexts, pelvic anatomy and loading are additionally shaped by parturition-related functional demands. Pelvic form varies substantially between individuals and between sexes, including dimensions that relate to inlet, midplane, and outlet architecture (Ruff, 2017; Weiss, 2020). Such variation can shift lever arms, articular orientation, and ligament attachment geometry,

and therefore may alter how otherwise standardized external forces are redistributed into SIJ micromotion. Beyond sex, skeletal tissues remodel across the lifespan, suggesting that age-related changes in geometry and material behavior may further modulate joint-level response (Farr and Khosla, 2015).

Finite-element (FE) modelling provides a complementary route to SIJ biomechanics because it can combine subject-specific anatomy with controlled boundary conditions and precisely defined load cases. Computational childbirth models highlight both the promise and the complexity of obstetric pelvic mechanics, including the interplay of geometry, tissue properties, and load direction, but many models are single-subject or focus on contact-rich scenarios rather than population-scale joint micromotion (Chen and Grimm, 2021; Roque de Almeida et al., 2024). More generally, SIJ FE studies are often limited by cohort size, by geometric normalization that removes absolute size information, or by uncertainty about whether between-subject dispersion is driven primarily by geometry (shape) versus material mapping.

Increasingly, large-cohort FE workflows adopt a common reference domain and encode subject-specific anatomy through an implicit mapping rather than explicit per-subject remeshing. This representation improves computational scalability, but it also shifts how “geometry” is represented and how size-related covariates should be interpreted: individual shape is carried by the mapping, even when the computational mesh is shared.

Two gaps are especially important for interpreting cohort-scale SIJ micromotion. First, variance in predicted motion can arise from at least two mechanistic “channels”: geometric variability (including joint orientation and pelvic ring architecture) and material variability (e.g., density-based bone stiffness mapping) (Keller, 1994). Distinguishing these channels is necessary to determine whether anatomy alone largely governs inter-subject variability under standardized loads, or whether plausible material heterogeneity contributes comparably. Second, size scaling of SIJ micromotion remains underreported in large cohorts. Under force-standardized loading, larger load-bearing geometry can reduce micromotion, and global size can confound sex comparisons given sexual dimorphism in pelvic morphology (Ruff, 2017; Weiss, 2020). In reference-domain workflows, this requires a consistent definition of “size” derived from the same implicit geometry representation as the FE solve.

To address these issues, we analyzed a large precomputed FE repository of 278 pelves in which subject geometry is represented on a common template through an implicit displacement mapping ϕ . The repository includes full, shape-only, and material-only variants, enabling

variance-channel decomposition under identical standardized load definitions. We evaluated SIJ rotations and translations under five static load cases spanning locomotor and parturition-motivated proxies (two-leg standing, one-leg standing, and three labor-motivated loading states). By combining directional (AP/ML/CC) outcomes with robust and cluster-robust regression, we aimed to quantify load-dependent directional re-routing, sex effects under parturition-motivated loading after anatomical adjustment, the relative contribution of shape versus material variance channels, and power-law size scaling of SIJ micromotion using total pelvic volume as a global proxy. We therefore tested the following pre-specified hypotheses.

Hypotheses

We pre-specified four mechanistic hypotheses, each phrased as an explicit, testable prediction:

1. **H1 (load dependence and re-routing):** under identical standardized resultant forces, SIJ micromotion exhibits load-case-specific patterns. Compared with bilateral standing, unilateral standing increases overall motion magnitude, whereas parturition-motivated loading produces non-uniform changes across AP/ML/CC translation components (directional re-routing) rather than uniform scaling of all components.
2. **H2 (sex effects conditional on anatomy):** under parturition-motivated loads, female sex is associated with greater SIJ micromotion after adjustment for age, a global size proxy, and key pelvic dimensions, consistent with sex-dimorphic pelvic architecture shaping compliance pathways.
3. **H3 (geometry-dominant variability):** between-subject variability in SIJ outcomes is dominated by geometric differences: shape-only simulations recover most of the full-model variance, whereas material-only perturbations explain a comparatively small fraction under the standardized loading protocol.
4. **H4 (size scaling):** SIJ micromotion follows load-dependent power-law scaling with global pelvic size. Under force-standardized standing loads, scaling is expected to be predominantly negative (larger pelves show smaller micromotion), and potential sex differences in scaling are evaluated via an interaction term rather than assumed a priori.

2 Materials and methods

2.1 Repository and cohort

We used precomputed outputs from a finite-element pelvic repository and two matched auxiliary variants designed to isolate shape-only and material-only effects. The cohort included 278 subjects (150 female, 128 male). Demographic metadata (age, sex), pelvic morphology (eight landmark-based dimensions), SIJ kinematics, and geometry-derived global metrics (volume, surface area, scale factor) were provided as structured tables accompanying the repository. All analyses were scripted and outputs were generated directly from these tables without manual value editing.

The morphology table contained eight pelvic dimensions: AP diameter, biacetabular width, biischadic width, bituberous width, iliopectineal eminence width, pelvic inlet transverse (PIT), sacral width, and subpubic angle. Missingness for demographic, morphometric, global size-proxy, and SIJ primary outcomes was zero.

Ethical considerations

This study analyzed a previously published, anonymized imaging-derived dataset and its associated precomputed finite-element repository (BoneDat; (Henyš and Kuchař, 2025)). No new human imaging or interventions were performed for this work. According to the BoneDat descriptor, use of the source imaging data was approved by the Ethics Committee of the University Hospital Hradec Králové (reference 202411IO3P) with waiver of informed consent; we refer readers to the BoneDat publication for full ethical and governance details.

Table 1: Cohort characteristics and pelvic dimensions. Values are mean $\pm$ SD unless noted.

Variable	Overall	Female	Male	Missing
N	278	150	128	0
Age (years)	53.90 $\pm$ 19.33	53.36 $\pm$ 19.74	54.52 $\pm$ 18.91	0
Inlet AP diameter (mm)	117.04 $\pm$ 10.52	121.04 $\pm$ 10.02	112.34 $\pm$ 9.07	0
Biacetabular width (mm)	125.81 $\pm$ 8.53	129.08 $\pm$ 8.21	121.98 $\pm$ 7.22	0
Biischadic width (mm)	104.27 $\pm$ 9.33	109.87 $\pm$ 7.13	97.71 $\pm$ 7.06	0
Bituberous width (mm)	132.24 $\pm$ 9.15	136.46 $\pm$ 8.29	127.30 $\pm$ 7.49	0
Iliopectineal eminence width (mm)	102.38 $\pm$ 7.75	106.20 $\pm$ 7.02	97.90 $\pm$ 5.96	0
Pelvic inlet transverse (mm)	131.58 $\pm$ 8.88	134.85 $\pm$ 8.59	127.75 $\pm$ 7.61	0
Sacral width (mm)	117.08 $\pm$ 6.08	116.75 $\pm$ 5.67	117.47 $\pm$ 6.54	0
Subpubic angle (deg)	78.12 $\pm$ 10.12	85.80 $\pm$ 6.14	69.11 $\pm$ 5.26	0
Global scale factor (-)	0.96 $\pm$ 0.06	0.92 $\pm$ 0.05	1.00 $\pm$ 0.05	0
Total pelvic volume (cm ³)	823.58 $\pm$ 149.32	736.79 $\pm$ 105.99	925.28 $\pm$ 126.99	0
Total pelvic surface (cm ²)	1568.63 $\pm$ 169.26	1479.09 $\pm$ 133.64	1673.57 $\pm$ 144.80	0

2.2 Geometry, material mapping, and boundary conditions

The FE workflow uses a small-strain isotropic elastic formulation posed on a common reference mesh. Subject-specific anatomy is introduced implicitly through a displacement mapping $\phi(\mathbf{X})$ that maps reference coordinates $\mathbf{X}$ to subject coordinates $\mathbf{x} = \mathbf{X} + \phi(\mathbf{X})$. This mapping is used in the elastic operators and to pull back loads defined on subject surfaces to the reference domain (Supporting Information: Supplementary Methods). Global size proxies (total volume, surface area, and a derived scale factor) were computed from the mapped geometry and provided alongside the kinematic outputs.

Material assignment followed repository defaults: bone modulus from density via a piecewise density–modulus relationship (Keller, 1994),

$$E = \begin{cases} 2398 \text{ MPa}, & \rho_{ash} \leq 0.486 \\ 10200 \rho_{ash}^2, & \rho_{ash} > 0.486 \end{cases}, \quad \rho_{ash} = \frac{\rho + 0.09}{1.14}, \quad (1)$$

with bone Poisson ratio 0.3. SIJ cartilage and pubic symphysis regions used modulus 1.0 MPa and Poisson ratio 0.45. Shape and density fields were mapped with radial basis interpolation (smoothing 50, 10 neighbors). Boundary restraint was imposed on the S1 facet using penalty enforcement (surface penalty $\gamma = 10^6$).

2.3 Load cases

Five standardized static load cases were analyzed: SP2leg, SP1leg, LAB1, LAB2, LAB3 (Table 2). Forces were applied as total-force vectors on named anatomical surfaces, scaled by loaded-surface area in each subject so that resultant force vectors were identical across subjects.

Table 2: Standardized load-case definitions used for SIJ analysis.

Load case	Mechanical proxy	Applied total forces	Resultant
SP2leg	Bilateral standing	[0,0,+400] N on right AC notch and [0,0,+400] N on left AC notch	800 N cranio-caudal
SP1leg	Unilateral standing	[0,0,+800] N on right AC notch	800 N cranio-caudal
LAB1	Mediolateral ring compression proxy	[+400,0,0] N on ring_contact_left and [-400,0,0] N on ring_contact_right	0 N net; bilateral ML pair
LAB2	Ischial loading / SIJ distraction proxy	[+400,0,0] N on left_ischium_tuber and [-400,0,0] N on right_ischium_tuber	0 N net; bilateral ML pair
LAB3	Outlet AP distraction proxy	[0,-400,0] N on pubis_ins and [0,+400,0] N on SCJ	0 N net; AP pair

2.4 SIJ kinematic outcomes

For each load, SIJ outputs were available for left and right joints as three rotations (nutation, AP-axis rotation, CC-axis rotation; degrees) and three translations (ML, AP, CC; mm). Primary

outcomes were:

1. mean left-right 3D rotation magnitude (deg),
2. mean left-right 3D translation magnitude (mm).

Secondary outcomes were absolute AP/ML/CC translation components. Additional derived outputs were rotational components and left-right asymmetry magnitudes. Mathematical and algorithmic details of the implicit geometry mapping, load pullback, SIJ rigid-body extraction from deformed geometry, derived scalar outcomes, and size-scaling exponent estimation are provided in the Supporting Information (Supplementary Methods).

2.5 Statistical models

Load-profile and directional re-routing (H1)

We summarized outcomes by load using medians and interquartile ranges. For directional re-routing, we computed within-subject differences relative to SP2leg for AP/ML/CC translation components and tested median shifts with Wilcoxon signed-rank tests; 95% CIs were bootstrap percentile intervals.

Sex effects under LAB loads (H2)

For each LAB phase and each primary outcome, we fit robust linear models (HC3):

$$Y \sim \text{sex} + \text{age}_z + \log(\text{volume})_z + \text{AP}_z + \text{Biischadic}_z + \text{Subpubic}_z + \text{sex} \times \log(\text{volume})_z. \quad (2)$$

The sex coefficient (female minus male) was reported with 95% CI, standardized coefficient, partial R^2 , and BH-FDR corrected p .

Pooled robust models across load cases (H1/H2/H4)

Because each subject contributed all five loads, we used cluster-robust OLS (clustered by subject ID) for each primary outcome:

$$Y \sim \text{load case} + \text{sex} + \text{age}_z + \log(\text{volume})_z + \text{AP}_z + \text{Biischadic}_z + \text{Subpubic}_z \quad (3)$$

$$+ \text{load case} \times \log(\text{volume})_z + \text{sex} \times \log(\text{volume})_z. \quad (4)$$

This model quantified load contrasts against SP2leg, adjusted sex effects, and load-dependent allometric modulation.

Variance-channel decomposition (H3)

For each load and SIJ metric, variance was computed in three matched simulation repositories: full, shape-only, and material-only. In the shape-only repository, the subject mapping ϕ was applied while the material mapping was held at the reference assignment; in the material-only repository, subject-specific material mapping was applied on the reference geometry (i.e., $\phi = 0$). We reported shape/full and material/full variance ratios (%) with bootstrap 95% CIs; because these are ratios of sample variances from separate simulation variants, values slightly above 100% can occur and indicate near-equality rather than literal “more than full” variance.

Size-scaling models (H4, Variant B)

For each load and each primary outcome, we fit robust log–log models:

$$\log(Y) \sim \log(\text{total volume}) + \text{sex} + \text{age}_z + \text{AP}_z + \text{Biischadic}_z + \text{Subpubic}_z + \log(\text{total volume}) \times \text{sex}. \quad (5)$$

Male and female scaling exponents were extracted from main and interaction terms. A supplementary sensitivity analysis replaced $\log(\text{total volume})$ with $\log(\text{scale})$.

2.6 Multiplicity and software

False-discovery control used Benjamini–Hochberg correction within each analysis family. We report corrected p values, 95% CIs, standardized betas, and partial R^2 . Analyses were run in Python (pandas, statsmodels, scipy, matplotlib, zarr).

3 Results

3.1 Cohort completeness and baseline morphology

The analyzed cohort comprised 278 subjects (150 female, 128 male), with age 53.9 ± 19.3 years (range 16–91). No missing values were present for demographic variables, pelvic dimensions, SIJ primary outcomes, or global size-proxy metrics. Female pelves showed larger mean inlet AP

diameter (121.0 vs 112.3 mm), biischadic width (109.9 vs 97.7 mm), and subpubic angle (85.8 vs 69.1°) than male pelves (Table 1).

3.2 Load-case dependence of SIJ micromotion (H1)

Median SIJ micromotion varied strongly by load (Table 3; Fig. 1). SP1leg produced the largest response (rotation 2.516 [1.981, 3.137] deg; translation 1.755 [1.458, 2.053] mm), exceeding SP2leg (1.692 [1.211, 2.451] deg; 1.299 [0.971, 1.625] mm). Relative to SP2leg, LAB1 and LAB2 reduced overall translation magnitude (0.665 and 0.764 mm medians), while LAB3 showed higher rotation than LAB1/LAB2 (1.769 deg) with low translation (0.790 mm median).

Cluster-robust pooled models confirmed these contrasts. For rotation magnitude, coefficients versus SP2leg were +0.723 deg for SP1leg, -0.326 deg for LAB1, -0.261 deg for LAB2, and +0.060 deg for LAB3 (all BH-FDR < 0.001). For translation magnitude, coefficients were +0.499 mm (SP1leg), -0.598 mm (LAB1), -0.498 mm (LAB2), and -0.494 mm (LAB3) (all BH-FDR < 10⁻²⁰).

Table 3: SIJ micromotion summary by load case. Values are median [IQR].

Load case	N	Rotation magnitude (deg)	Translation magnitude (mm)	AP translation component (mm)	ML translation component (mm)	CC translation component (mm)
SP2leg	278	1.692 [1.211, 2.451]	1.299 [0.971, 1.625]	0.272 [0.158, 0.433]	0.318 [0.160, 0.574]	1.131 [0.855, 1.459]
SP1leg	278	2.516 [1.981, 3.137]	1.755 [1.458, 2.053]	0.431 [0.299, 0.572]	0.341 [0.200, 0.550]	1.527 [1.262, 1.775]
LAB1	278	1.393 [0.908, 2.060]	0.665 [0.450, 0.923]	0.268 [0.169, 0.424]	0.410 [0.228, 0.678]	0.263 [0.163, 0.410]
LAB2	278	1.433 [1.011, 2.069]	0.764 [0.536, 1.048]	0.413 [0.216, 0.628]	0.476 [0.284, 0.738]	0.236 [0.138, 0.377]
LAB3	278	1.769 [1.169, 2.591]	0.790 [0.568, 1.052]	0.262 [0.143, 0.473]	0.274 [0.138, 0.511]	0.534 [0.325, 0.749]

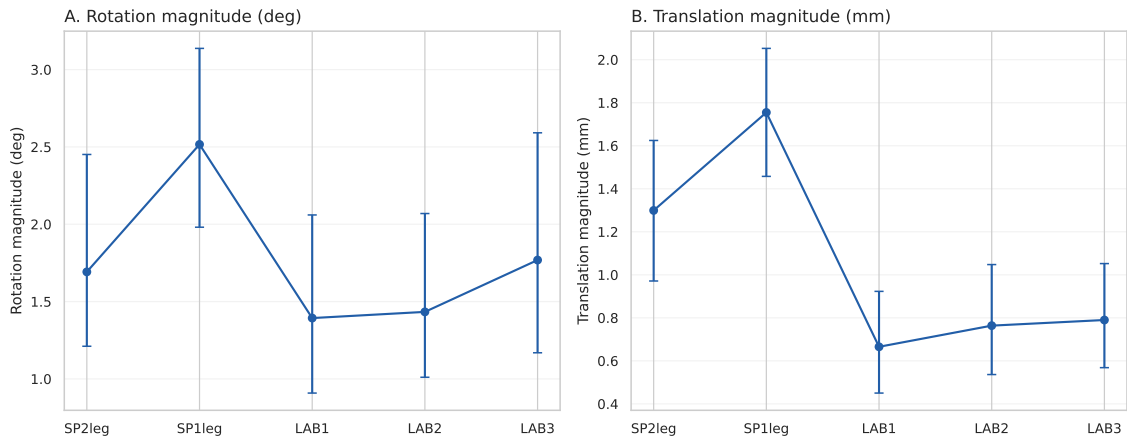


Figure 1: Primary SIJ outcomes across load cases (medians with IQR error bars).

3.3 Directional re-routing rather than uniform scaling (H1)

Directional translation differences versus SP2leg showed load-specific anisotropy (Fig. 2; Supplementary Table S1). SP1leg increased AP (+0.072 mm, 95% CI 0.045 to 0.096, BH-FDR 1.31×10^{-22}) and CC (+0.394 mm, 0.347 to 0.434, BH-FDR 4.73×10^{-43}) components, while ML change was small and not significant after correction.

LAB loads exhibited clear re-routing. LAB1 increased ML (+0.086 mm, 0.077 to 0.095) but strongly reduced CC (-0.850 mm, -0.881 to -0.811), with no AP change after correction. LAB2 increased AP (+0.163 mm, 0.102 to 0.213) and ML (+0.145 mm, 0.132 to 0.160), with strong CC reduction (-0.903 mm, -0.931 to -0.853). LAB3 showed reduced ML (-0.041 mm, -0.052 to -0.033) and reduced CC (-0.586 mm, -0.609 to -0.544), while AP remained unchanged.

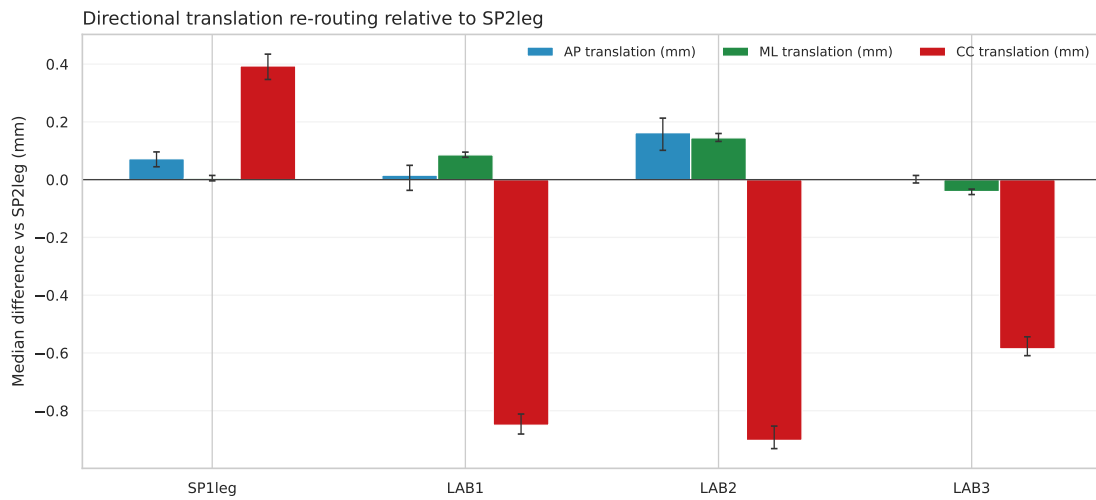


Figure 2: Directional translation re-routing relative to SP2leg. Bars show median within-subject differences with 95% bootstrap CIs.

3.4 Sex effects under parturition-motivated loads (H2)

Adjusted LAB-phase robust models showed higher female micromotion for most primary outcomes (Fig. 3; Supplementary Table S3). For translation magnitude, female effects were +0.229 mm in LAB1 (95% CI 0.078 to 0.379; BH-FDR 0.007), +0.199 mm in LAB2 (0.039 to 0.359; BH-FDR 0.022), and +0.233 mm in LAB3 (0.078 to 0.388; BH-FDR 0.007). For rotation magnitude, the female effect was strongest in LAB3 (+0.627 deg, 0.206 to 1.048; BH-FDR 0.007); LAB1 was borderline after correction (BH-FDR 0.054).

In pooled cluster-robust models across all loads, female sex remained associated with higher rotation (+0.450 deg, BH-FDR 0.020) and translation (+0.254 mm, BH-FDR 3.41×10^{-4}).

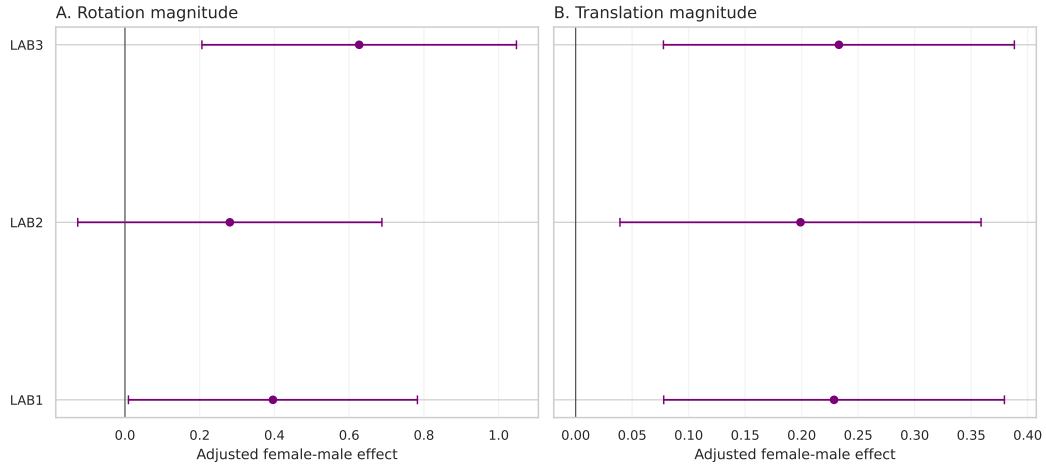


Figure 3: Adjusted female-minus-male effects for primary outcomes in LAB1–LAB3 (95% CI).

3.5 Variance channels: shape-dominant variability (H3)

Variance-channel comparison indicated geometry-dominant variability. Shape-only simulations reproduced full-model variance to within a narrow margin (98.7–102.1% across load/metric combinations; median 100.46%), while material-only variability was orders of magnitude smaller (0.001–0.317%; median 0.033%) (Fig. 4; Supplementary Table S2). Ratios slightly above 100% can arise because these are sample-variance ratios computed on separate simulation variants (and because material mapping can partially compensate for geometric effects); values near 100% therefore indicate near-complete retention rather than a paradoxical “excess” of variance.

For LAB phases specifically, shape/full ratios were close to 100% for all primary and directional metrics, while material/full remained below 0.28% in all LAB combinations. This supports dominant geometry-driven inter-subject SIJ variability under standardized loading.

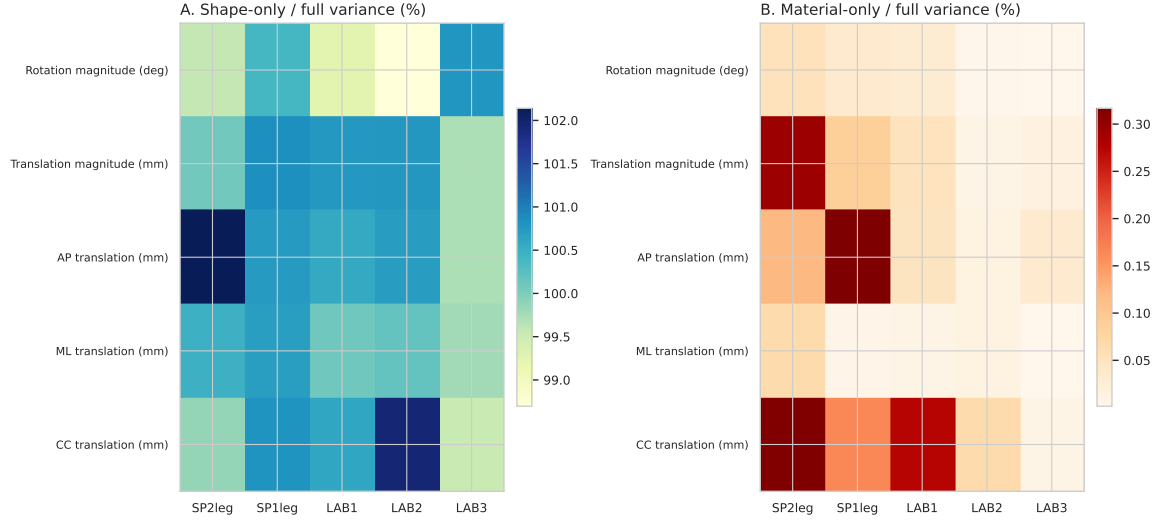


Figure 4: Variance-channel ratios across load cases and SIJ metrics.

3.6 Scaling of SIJ micromotion with global pelvic size (H4)

Load-wise log-log allometry models are summarized in Table 4. Under SP1leg, larger pelvic volume was associated with lower micromotion in both sexes: rotation exponents were -0.482 (male, 95% CI -0.870 to -0.094) and -0.428 (female, -0.762 to -0.093); translation exponents were -0.569 (male, -0.869 to -0.269) and -0.432 (female, -0.669 to -0.194). Several LAB-phase exponents had CIs crossing zero.

Sex-by-size interaction terms for allometry slopes were not significant after BH-FDR correction in any load/outcome pair (all corrected $p \geq 0.958$), indicating no strong evidence for sex-divergent scaling exponents in this dataset.

In pooled cluster-robust models, translation magnitude showed an overall negative size association (log-total-volume coefficient -0.092, 95% CI -0.143 to -0.041; BH-FDR 6.17×10^{-4}), with significant load-dependent modulation through load $\times$ size interactions.

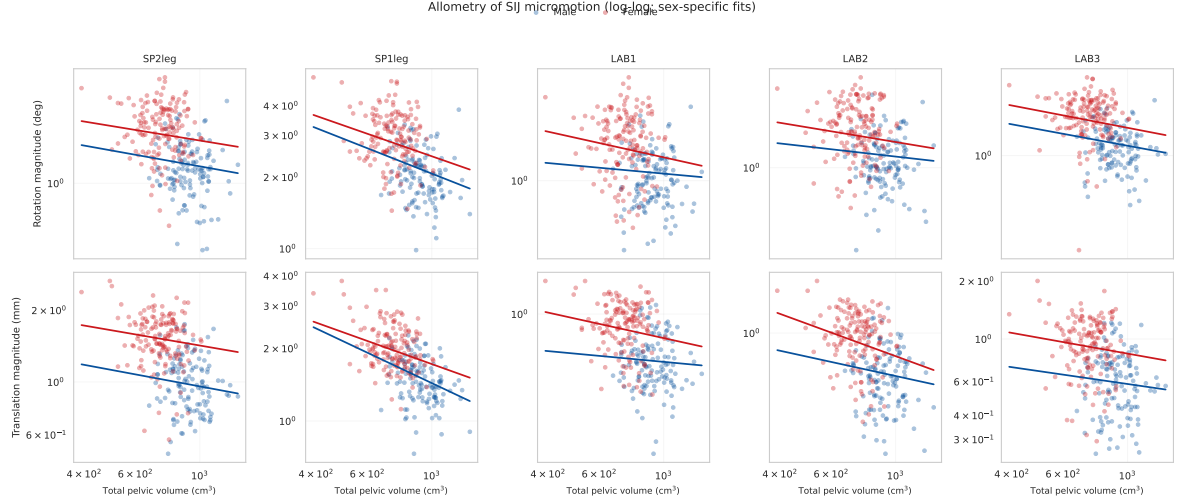


Figure 5: Log-log allometry of primary SIJ outcomes versus total pelvic volume, stratified by sex and load case.

Table 4: Primary robust regression and size-scaling results. Panel A: pooled cluster-robust primary models. Panel B: load-wise scaling exponents.

Panel A

outcome	term	beta	ci95_low	ci95_high	beta_std	partial_r2	p_fdr_bh
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]	0.723	0.672	0.773	0.781	0.362	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]	-0.326	-0.359	-0.292	-0.352	0.209	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]	-0.261	-0.323	-0.198	-0.282	0.046	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]	0.060	0.033	0.087	0.065	0.013	0.000
rot_mag_deg	sex_F	0.450	0.103	0.797	0.487	0.005	0.020
rot_mag_deg	age_z	0.044	-0.053	0.141	0.048	0.001	0.398
rot_mag_deg	log_total_volume_z	-0.126	-0.285	0.034	-0.136	0.002	0.182
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]:log_total_volume_z	-0.032	-0.079	0.015	-0.035	0.001	0.223
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]:log_total_volume_z	0.096	0.062	0.130	0.104	0.022	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]:log_total_volume_z	0.105	0.033	0.177	0.114	0.006	0.009
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]:log_total_volume_z	-0.065	-0.093	-0.038	-0.071	0.016	0.000
rot_mag_deg	AP_z	0.130	0.028	0.232	0.141	0.005	0.020
rot_mag_deg	BiischadicWidth_z	0.082	-0.035	0.200	0.089	0.001	0.223
rot_mag_deg	SubpubicAngle_z	0.079	-0.075	0.232	0.085	0.001	0.366
rot_mag_deg	sex_F:log_total_volume_z	-0.079	-0.280	0.123	-0.085	0.000	0.443
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]	0.499	0.465	0.533	0.884	0.375	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]	-0.598	-0.627	-0.569	-1.060	0.542	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]	-0.498	-0.533	-0.462	-0.882	0.353	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]	-0.494	-0.513	-0.475	-0.875	0.659	0.000
trans_mag_mm	sex_F	0.254	0.121	0.387	0.450	0.010	0.000
trans_mag_mm	age_z	0.028	-0.008	0.063	0.049	0.002	0.134
trans_mag_mm	log_total_volume_z	-0.092	-0.143	-0.041	-0.163	0.009	0.001
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]:log_total_volume_z	-0.073	-0.106	-0.040	-0.129	0.013	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]:log_total_volume_z	0.073	0.043	0.103	0.130	0.016	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]:log_total_volume_z	0.048	0.008	0.088	0.085	0.004	0.025
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]:log_total_volume_z	0.085	0.068	0.102	0.150	0.063	0.000
trans_mag_mm	AP_z	-0.062	-0.095	-0.028	-0.109	0.009	0.000
trans_mag_mm	BiischadicWidth_z	0.043	-0.002	0.089	0.076	0.002	0.079
trans_mag_mm	SubpubicAngle_z	0.051	-0.012	0.114	0.091	0.002	0.129
trans_mag_mm	sex_F:log_total_volume_z	-0.039	-0.111	0.034	-0.069	0.001	0.293

Panel B

load_case	outcome	male_exponent	male_ci95_low	male_ci95_high	female_exponent	female_ci95_low	female_ci95_high	sex_slope_interaction_beta	interaction_p_fdr_bh	adj_r2
SP2leg	rot_mag_deg	-0.334	-0.931	0.263	-0.306	-0.762	0.151	0.028	0.958	0.383
SP2leg	trans_mag_mm	-0.227	-0.562	0.107	-0.211	-0.505	0.084	0.017	0.958	0.468
SP1leg	rot_mag_deg	-0.482	-0.870	-0.094	-0.428	-0.762	-0.093	0.055	0.958	0.382
SP1leg	trans_mag_mm	-0.569	-0.869	-0.269	-0.432	-0.669	-0.194	0.137	0.958	0.500
LAB_phase1	rot_mag_deg	-0.172	-0.846	0.502	-0.416	-0.959	0.127	-0.244	0.958	0.267
LAB_phase1	trans_mag_mm	-0.177	-0.665	0.311	-0.419	-0.914	0.077	-0.242	0.958	0.354
LAB_phase2	rot_mag_deg	-0.228	-1.020	0.565	-0.334	-0.891	0.223	-0.106	0.958	0.269
LAB_phase2	trans_mag_mm	-0.367	-0.904	0.171	-0.614	-1.061	-0.166	-0.247	0.958	0.375
LAB_phase3	rot_mag_deg	-0.498	-1.169	0.174	-0.518	-1.048	0.011	-0.021	0.958	0.367
LAB_phase3	trans_mag_mm	-0.220	-0.720	0.280	-0.271	-0.708	0.166	-0.051	0.958	0.310

3.7 Summary relative to hypotheses

Taken together, the results provide the following hypothesis-level interpretation. **H1** was supported: SIJ micromotion differed across load cases, and parturition-motivated loads produced directional re-routing rather than uniform scaling (Table 3; Figs. 1–2). **H2** was supported overall, with female-associated increases in adjusted LAB-phase micromotion, most consistently for translation magnitude (Fig. 3). **H3** was supported: between-subject variability was largely reproduced by shape-only simulations, while material-only fractions were small (Fig. 4). **H4** was partially supported: size scaling was load dependent with predominantly negative scaling under standing loads, but sex-by-size slope differences were not supported after multiplicity control (Fig. 5; Table 4).

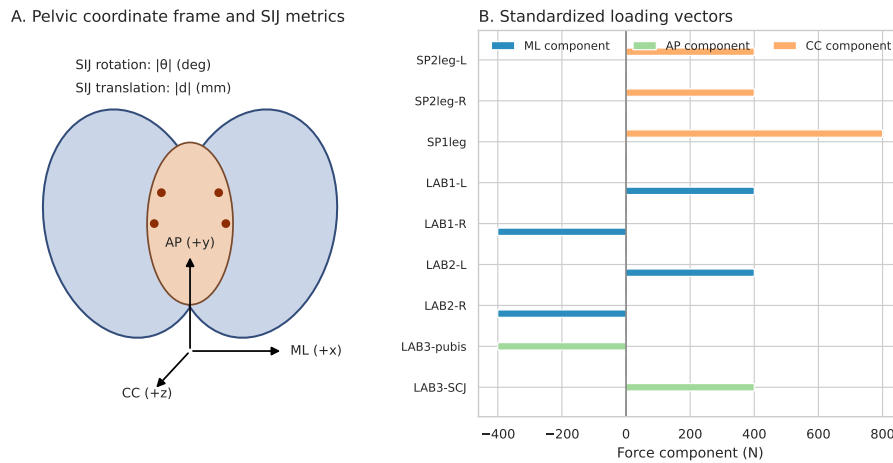


Figure 6: Anatomy schematic, pelvic coordinate definition, SIJ metric definitions, and standardized load vectors.

4 Discussion

This study used a large, precomputed FE repository to characterize SIJ micromotion under standardized standing and parturition-motivated loading with explicit anatomy and global size-proxy covariates derived from an implicit geometry mapping. The analysis was structured around four pre-specified hypotheses (H1–H4) concerning load dependence and directional re-routing, sex effects under parturition-motivated loading, variance channels, and size scaling.

Interpretation by hypothesis

H1 (load dependence and re-routing). SIJ response was strongly load dependent and direction-specific. One-leg standing generated the largest overall SIJ micromotion, consistent with unilateral support demands and larger asymmetric hip loading reported in instrumented and musculoskeletal literature (Bergmann et al., 2001; Kutzner et al., 2016). In contrast, the parturition-motivated load proxies produced lower global translation magnitude than standing but clear directional re-routing: LAB1 and LAB2 shifted motion toward ML and AP components while markedly reducing CC translation, and LAB3 preserved CC reduction with a distinct ML pattern. These observations support an anisotropic interpretation of SIJ mechanics in which load direction changes the partitioning of motion across AP/ML/CC channels rather than scaling micromotion uniformly (Vleeming et al., 2012; Goode et al., 2008; Meijer et al., 2014).

H2 (sex effects conditional on anatomy). Under parturition-motivated loading, female sex was associated with higher adjusted SIJ micromotion, most consistently for translation magnitude across LAB phases and for rotation magnitude in the LAB3 proxy. Because these effects persisted after adjustment for age, a global size proxy, and key pelvic dimensions, they are unlikely to be explained solely by overall size. Instead, they are consistent with sex-dimorphic pelvic architecture shaping compliance pathways and load routing through the pelvic ring (Ruff, 2017; Weiss, 2020). At the same time, the adjustment set captures only a subset of potentially relevant anatomical features (e.g., joint surface orientation and ligament attachment geometry), so the sex coefficient should be interpreted as a summary association rather than a single-cause mechanism.

H3 (geometry-dominant variability). Between-subject variability in SIJ outcomes was geometry-dominant. Shape-only outputs reproduced essentially all full-model variance, whereas material-only variance fractions were very small. Values slightly above 100% for the shape/full ratio are not paradoxical: they reflect ratios of sample variances computed on separate simulation variants, and material mapping can partially compensate geometric effects in the full model. Under the present force-standardized loading protocol, these findings indicate that subject-specific morphology and geometry carry most of the inter-subject dispersion signal, while material-only perturbation contributes comparatively little to spread in the tested outcomes.

H4 (size scaling). Size scaling was load dependent and generally negative for several standing outcomes, meaning larger pelves tended to show smaller SIJ micromotion under identical resultant forces. This is mechanically plausible in a force-standardized setting: larger load-bearing geometry can distribute identical resultant loads over broader structural pathways. Sex-specific differences in scaling slope were not supported after multiplicity correction, suggesting broadly similar scaling of micromotion with global size across sexes in this dataset.

Anatomical and functional implications

The combined findings support an anatomy-first interpretation of SIJ mechanics: load direction and pelvic geometry govern how micromotion is partitioned across AP/ML/CC channels. From an obstetric-mechanical perspective, this suggests that evaluating only scalar mobility may miss clinically relevant directional behavior, particularly under parturition-motivated loading states. The observed sex differences in adjusted micromotion, together with known sex contrasts in pelvic morphology (Ruff, 2017; Weiss, 2020), support the view that group-level differences emerge from load-case-specific routing through a sexually dimorphic pelvic ring rather than from a uniform “more mobile” versus “less mobile” SIJ phenotype.

Limitations

The analysis remains constrained by the assumptions of the underlying repository. The mechanical model is linear, small-strain, and static; it does not include time-varying contractions, contact-rich fetal-head interactions, or hormonally mediated tissue adaptation (Chen and Grimm, 2021; Roque de Almeida et al., 2024). The LAB load cases are therefore proxies, not full labor reconstructions. Material mapping follows a fixed density–modulus law and fixed cartilage/symphysis parameters. In addition, although age was adjusted statistically, the model does not explicitly encode age-dependent tissue remodeling trajectories (Farr and Khosla, 2015). These boundaries should temper causal interpretation.

Outlook

Future extensions should combine this population-scale geometry framework with nonlinear contact models and subject-specific soft-tissue updates across pregnancy. A second priority is external validation against in-vivo SIJ kinematic datasets and imaging-derived outlet/inlet deformation measurements in matched loading scenarios.

5 Conclusion

In a 278-subject pelvic FE cohort represented on a common template via an implicit geometry mapping, SIJ micromotion under standardized loads showed strong load dependence, directional re-routing under parturition-motivated proxies, and dominant geometry-driven variance. Female-associated increases in LAB-phase micromotion persisted after adjustment for age, a global size proxy, and pelvic dimensions. Size scaling indicated mainly negative scaling of SIJ micromotion with pelvic volume in several load contexts, without robust evidence for sex-divergent scaling slopes. Together, these results support an anatomy-centered view of SIJ function in which geometry and load direction are primary determinants of micromotion patterns.

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Author contributions

CRedit authorship contribution statement. Petr Henyš: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Funding acquisition; Writing – original draft; Writing – review & editing. Niels Hammer: Supervision; Writing – review & editing. All authors reviewed and approved the final manuscript.

Conflict of interest statement

The authors declare no conflicts of interest.

Supporting information

Supporting Information includes Supplementary Methods (implicit mapping, load pullback, SIJ kinematics, and size-scaling model details) and Supplementary Tables S1–S4.

Data Availability Statement

The derived data supporting the findings of this study (summary tables and figures) are provided with the manuscript. Subject-level derived datasets and analysis outputs are available in a blinded repository for peer review and will be made publicly available upon publication.

Code Availability Statement

All custom code used to post-process the finite-element outputs and perform the statistical analyses is available in the same blinded repository for peer review and will be made publicly available upon publication.

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